Detecting Misinterpreted but Genuine Citations through Automated Claim Verification

A Review and Synthesis of Natural Language Processing Approaches to Citation Integrity

Abstract

Citations are the connective tissue of scholarly communication, allowing claims to be traced back to supporting evidence. A growing body of evidence shows that many citations, while pointing to real and legitimately published sources, nonetheless misrepresent what those sources actually say — a problem distinct from citation fabrication or "hallucination," in which the cited work does not exist at all. This paper surveys the emerging natural language processing (NLP) literature on detecting such *misinterpreted but genuine citations* through automated claim verification. We review empirical studies documenting the prevalence and consequences of citation inaccuracy, position the problem within the broader taxonomy of citation distortion (bias, amplification, and invention), and examine how techniques originally developed for general-purpose fact verification — including entailment-based claim checking, retrieval-augmented evidence extraction, and large language model (LLM) prompting — have been adapted to the scholarly domain. We synthesize findings from recent corpus-based studies and replication experiments, showing that automated systems perform well at coarse-grained tasks such as stance detection and citation amplification but struggle with the subtler forms of citation invention, where distinguishing an accurate paraphrase from a genuine misrepresentation requires deep contextual and domain understanding. We conclude by identifying research gaps — including full-text access barriers, weak generalization across disciplines, and a persistent tendency of automated systems toward overprediction — and propose directions for building more reliable, explainable, and scalable citation-verification pipelines suitable for integration into peer review and scholarly publishing workflows.

Keywords: citation accuracy; citation distortion; quotation errors; scientific claim verification; natural language processing; large language models; misinformation; scholarly communication; citation integrity

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1 Introduction

Citation is the principal mechanism by which scientific claims accumulate, and are checked against, an evidentiary record. When a citing author correctly represents a source's findings, citation functions as intended: readers can trust that a claim is traceable to supporting data. When a citation is inaccurate — whether through carelessness, misreading, or motivated reasoning — the citing text can present an unsupported, exaggerated, or outright contradicted claim as though it were settled science. Ngatuvai et al. (2021) frame accurate citation as foundational to the credibility of clinical research, and reviews across medical specialties have repeatedly found that a substantial fraction of citations do not accurately reflect their sources (de Lacey et al., 1985; Mogull, 2017; Jergas & Baethge, 2015).

Two categories of citation problems are frequently conflated in public discussion but are analytically distinct. The first is *citation existence failure*: a reference that is fabricated, non-existent, or bibliographically corrupted — a phenomenon that has drawn intense recent attention because of its prevalence in text generated by large language models (Misra & Udandarao, 2026). The second, which is the focus of this paper, is *citation interpretation failure*: the cited work is real, correctly identified, and bibliographically intact, yet the citing sentence misrepresents, overstates, dilutes, or otherwise distorts what that work actually reports. This second category is arguably more insidious, because it cannot be caught by simply checking whether a reference resolves to a real document — the reference is genuine, but the claim attributed to it is not faithfully supported. Sarol et al. (2024) refer to these as *quotation errors*, and Greenberg (2009) documents how, aggregated across a citation network, such errors can manufacture "unfounded authority" for a claim that was never rigorously established.

This paper examines how automated claim verification — a family of NLP techniques originally developed for fact-checking in journalism and open-domain question answering (Thorne et al., 2018; Guo et al., 2022) — has been adapted to detect misinterpreted but genuine citations in the scientific literature. We review the empirical literature documenting the scope of the problem, describe the technical building blocks (citation context extraction, evidence retrieval, textual entailment, and stance/sentiment classification) that underlie current systems, summarize recent corpus-based and replication studies, and discuss the challenges that keep this from being a solved problem. The paper closes by identifying gaps in current research and proposing directions that could make automated citation verification a practical component of scholarly publishing infrastructure.

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2 Background, Related Work, and Literature Review

2.1 Empirical evidence of citation inaccuracy

Concern about citation fidelity predates modern NLP by decades. de Lacey et al. (1985) conducted one of the earliest systematic audits of quotation and reference accuracy in medical journals, finding that a meaningful share of cited "facts" did not match their sources. Since then, dozens of specialty-specific audits — in surgery, dermatology, neuroanesthesiology, and orthopedics, among others — have repeated this exercise, generally converging on the estimate that roughly 20–25% of quoted assertions in the medical literature are inaccurately cited (Mogull, 2017). Jergas and Baethge (2015) synthesized this body of work in a systematic review and meta-analysis of quotation accuracy in medical journal articles, formalizing "quotation error" as a term of art for citations whose asserted content is not actually supported by the cited source.

Mogull (2017) revisited the methodology behind these estimates and, using a stricter denominator (errors per quotation examined, rather than per reference selected), recalculated the rate of quotation errors at 14.5% (95% CI: 10.5–18.6%). Critically, this study also decomposed errors by severity: 64.8% of errors were "major" — the referenced source failed to substantiate, was unrelated to, or directly contradicted the citing assertion — while 35.2% were "minor," consisting of oversimplification, overgeneralization, or otherwise trivial inaccuracies. This distinction matters for automated detection: major errors, which more closely resemble outright contradiction or irrelevance, are plausibly easier for NLP systems to flag than minor errors, which require a finer-grained semantic judgment about degree of distortion.

2.2 Citation distortion at the network level

Individual quotation errors are troubling on their own, but their aggregate effect across a citation network can be more consequential. Greenberg (2009) provided an influential case study by reconstructing the complete citation network — 242 papers and 675 citations — surrounding the claim that β-amyloid, the protein implicated in Alzheimer's disease, is abnormally present in and damages the skeletal muscle of patients with inclusion body myositis. Despite weak underlying evidence (of twelve primary data papers, six were critical and six were supportive, yet the six supportive papers received 94% of all citations to primary data), the claim became broadly accepted in the literature. Greenberg attributed this to three interacting mechanisms, subsequently termed *citation distortions*: (1) **citation bias**, the disproportionate citation of supportive over critical primary evidence; (2) **citation amplification**, whereby a claim gains authority through repeated citation of secondary sources such as literature reviews rather than the primary data itself (in Greenberg's network, 95% of citation paths passed through just four review articles); and (3) **citation invention**, in which citing authors alter a claim's evidential basis — for example, citing a conference abstract as though it were peer-reviewed (termed "back door invention"), or converting a hypothesis into an assertion of fact through citation alone ("citation transmutation"). Ravnskov (1992) had earlier documented an analogous pattern of citation bias in trials of cholesterol-lowering therapy for coronary heart disease, in which supportive trials were cited markedly more often than trials with null or contrary findings — a concern that Ahn and Kang (2018) note is especially serious in systematic reviews and meta-analyses, which are expected to synthesize evidence without such asymmetry.

The real-world stakes of citation distortion are illustrated by the role citation practices are argued to have played in the opioid crisis. A single one-paragraph letter to the *New England Journal of Medicine* asserted, without rigorous supporting data, that addiction was rare among hospitalized patients treated with narcotics (Porter & Jick, 1980). Leung et al. (2017) traced how this brief, largely unsubstantiated letter was cited hundreds of times over subsequent decades — frequently in ways that mischaracterized its content as a definitive, generalizable finding about outpatient prescribing — contributing to a climate of opioid overprescription. This example demonstrates concretely why *misinterpreted but genuine* citations deserve independent scholarly attention: the letter was a real, identifiable publication, and its existence was never in question, but what many citing authors claimed it showed diverged substantially from its actual, narrow content.

2.3 Citation function, intent, and sentiment classification

Adjacent to citation accuracy is a longer-standing NLP tradition of classifying the rhetorical role a citation plays. Teufel, Siddharthan, and Tidhar (2006) introduced an influential annotation scheme and supervised classifier for *citation function* — the author's reason for citing a given work (e.g., using a method, contrasting results, acknowledging prior work) — built on shallow and linguistically informed features. This line of work was substantially advanced by Cohan, Ammar, van Zuylen, and Cady (2019), who introduced SciCite, a large dataset for *citation intent classification*, and a multitask neural model using "structural scaffolds" (auxiliary prediction of section title and citation-worthiness) that achieved a new state of the art on the existing ACL-ARC benchmark, improving F1 by 13.3 absolute percentage points. Related work has examined *citation sentiment* — whether a citation context expresses a positive, negative, or neutral stance toward the cited claim. Kilicoglu et al. (2019) developed a convolutional neural network for classifying citation sentiment in clinical research publications, while more general stance-detection resources, such as the tweet-based dataset of Mohammad et al. (2016) and the Fake News Challenge baseline of Riedel, Augenstein, Spithourakis, and Riedel (2017), have been repurposed for citation-stance tasks by treating a scientific claim as the "headline" and the citation context as the "article body" to classify. These citation-function, -intent, and -sentiment classifiers do not, by themselves, verify whether a citation is *accurate* — but as later sections show, they form essential building blocks for detecting bias and amplification, two of Greenberg's three distortion types.

2.4 Automated fact and scientific claim verification

The technical apparatus for detecting misinterpreted citations descends largely from automated fact-checking research. Thorne, Vlachos, Christodoulopoulos, and Mittal (2018) introduced FEVER, a large-scale dataset and shared task in which a system must retrieve evidence from Wikipedia and classify a claim as supported, refuted, or lacking sufficient evidence — establishing the now-standard three-stage pipeline of claim detection, evidence retrieval, and verification (formalized comprehensively in the survey by Guo, Schlichtkrull, and Vlachos, 2022). Krishna, Riedel, and Vlachos (2022) extended verification methodology with ProoFVer, a natural-logic theorem-proving approach that produces interpretable proof traces for veracity decisions.

Because general-domain fact-checking datasets are poorly matched to the specialized vocabulary and argumentation style of scientific writing, Wadden et al. (2020) introduced SciFact, motivated in part by the need to verify COVID-19-related claims, in which a claim must be checked against a corpus of scientific abstracts and, if evidence exists, the supporting or refuting "rationale" sentences must be identified. Wadden et al. (2022) subsequently developed MultiVerS, which improves on this task using weak supervision and full-document (rather than abstract-only) context, better reflecting the reality that supporting or contradicting evidence for a claim is often found deep in a paper's body text rather than its abstract. These scientific claim verification systems are a natural fit for citation checking: a citing sentence can be treated as the "claim" and the cited paper's full text as the evidence pool to be searched.

2.5 Distinguishing citation interpretation failure from citation existence failure

A parallel and increasingly visible strand of research addresses *citation hallucination* — references generated by LLMs that are fabricated, bibliographically corrupted, or otherwise non-existent. Misra and Udandarao (2026) report hallucination rates ranging from roughly 18% for GPT-4 to over 70% for some other frontier models, with domain-specific rates as high as 88% in legal contexts, and note that even purpose-built benchmarks such as CiteME show a wide gap between LLM accuracy (4.2–18.5%) and human annotators (69.7%) at verifying whether a citation is genuine. Tools built to address this problem — such as retrieval-grounded verifiers that check whether a cited work exists and whether its metadata (authors, year, title, venue) is correct — solve a categorically different problem from the one addressed in this paper. A citation can pass every existence and metadata check and still misrepresent its source; conversely, methods for verifying citation *interpretation* generally presuppose that the cited document is real and accessible. Keeping this distinction explicit is important both for research design (the two problems require different evidence sources — bibliographic databases versus full source text) and for downstream tool design, since a scholarly assistant that only checks reference existence provides no protection against the subtler and arguably more consequential problem of misquotation.

2.6 Adjacent scholarly-error detection

Citation verification sits within a broader family of efforts to automate the detection of scientific reporting errors. Nuijten and Polanin (2020) developed statcheck, a tool that automatically extracts reported statistical test results from articles and recomputes p-values to flag internal inconsistencies, and later work found that mandating its use in peer review measurably reduced reporting errors in participating journals. Meuschke et al. (2023) benchmarked the PDF-to-text extraction tools that many of these downstream error-detection pipelines depend on, finding that conversion artifacts remain a nontrivial source of noise — a finding directly relevant to citation verification systems that require clean full-text input from both the citing and cited documents.

3 Main Thematic and Technical Discussion

3.1 A working taxonomy of citation problems

Building on Greenberg (2009) and the operationalization introduced by Sarol, Ming, Radhakrishna, Schneider, and Kilicoglu (2024), it is useful to organize citation problems along two axes: (a) whether the problem concerns the *existence* of the cited source or the *fidelity* of the claim attributed to it, and (b) whether the problem is detectable from a single citing–cited pair or only emerges at the level of a citation *network*. Table 1 summarizes this taxonomy.

Table 1. A taxonomy of citation problems relevant to automated verification.

Problem type	Unit of analysis	Cited source genuine?	Representative work
Citation hallucination / fabrication	Single reference	No	Misra & Udandarao (2026)
Citation diversion (content reinterpreted)	Single citing–cited pair	Yes	Greenberg (2009); Sarol et al. (2024)
Citation transmutation (hypothesis → fact)	Single citing–cited pair	Yes	Greenberg (2009)
Dead-end citation (no relevant content)	Single citing–cited pair	Yes	Greenberg (2009); Sarol et al. (2024)
Back-door invention (abstract cited as peer-reviewed)	Single citing–cited pair	Yes	Greenberg (2009)
Title invention (title overstates body content)	Single document	Yes	Greenberg (2009); Sarol et al. (2025)
Citation bias (selective citation of supportive evidence)	Network	Yes	Greenberg (2009); Ravnskov (1992)
Citation amplification (claim inflated via secondary sources)	Network	Yes	Greenberg (2009)

The subject of this paper — misinterpreted but genuine citations — spans the middle rows of this table: diversion, transmutation, dead-end citation, and back-door invention are all *pairwise* phenomena in which a real citing sentence misrepresents a real cited source, while bias and amplification require reasoning over an entire citation network built from many such pairs.

3.2 Formalizing citation verification as claim verification

The technical move that unites most current approaches is to recast a citation as a claim-evidence verification problem, directly analogous to FEVER (Thorne et al., 2018) or SciFact (Wadden et al., 2020): given (i) a *citation context* — the sentence or short passage in the citing document that attributes content to a reference — and (ii) the *cited source's* full text (or abstract), determine whether the source substantiates, contradicts, or is simply irrelevant to the claim attributed to it. Sarol et al. (2024) instantiate this directly, framing citation accuracy classification as assigning each citation context one of three labels — **ACCURATE**, **INACCURATE** (the cited content is altered or misrepresented), or **IRRELEVANT** (the citation context describes content not actually present in the cited work) — closely paralleling FEVER's SUPPORTED/REFUTED/NOT

ENOUGH INFO scheme and directly operationalizing Greenberg's notions of citation diversion and dead-end citation, respectively.

A full verification pipeline typically involves four stages:

1. **Citation context extraction.** Locate the sentence(s) in the citing document associated with a given in-text citation marker, typically via regular-expression or rule-based parsing of reference call-outs, sometimes expanded to a surrounding window to capture elided context.
2. **Evidence retrieval.** Within the cited source, retrieve the passage(s) most relevant to the citation context. Because the entire full text of a source paper is usually too long to process directly, retrieval methods — ranging from simple lexical overlap (e.g., BM25) to neural re-ranking (e.g., MonoT5) — are used to narrow candidate evidence sentences before verification (Sarol et al., 2024).
3. **Stance or accuracy classification.** Given the citation context and the retrieved evidence, classify the relationship as supportive, contradictory, neutral, or absent, using either supervised classifiers trained on citation-specific corpora (Kilicoglu et al., 2019) or general-purpose scientific claim verification models such as MultiVerS (Wadden et al., 2022).
4. **Aggregation (optional).** For network-level phenomena such as bias and amplification, pairwise verification outputs are aggregated across all citations to a claim to detect asymmetries in supportive versus critical citation counts (Greenberg, 2009; Sarol et al., 2025).

3.3 Model architectures in current use

Two broad architectural families dominate current systems. The first is **non-LLM, task-specific pipelines**, which combine publication metadata (e.g., MeSH terms and publication type in biomedical databases), rule-based context extraction, and purpose-built classifiers — support vector machines or fine-tuned transformer encoders such as SciBERT — trained on annotated citation corpora (Sarol et al., 2024). The second is **LLM-based prompting**, in which a general-purpose model such as GPT-4 or GPT-4o is given the citation context, the cited full text (or retrieved excerpts), and a natural-language definition of the target distortion type, and asked to output a categorical judgment directly, without task-specific fine-tuning (Sarol et al., 2025; Zhang & Abernethy, 2024). Retrieval-augmented generation (Lewis et al., 2020) is increasingly used to combine the two: rather than feeding an LLM an entire cited document, a retrieval step first narrows the input to the passages most likely to be relevant, mitigating both context-window limitations and the tendency of long, noisy inputs to degrade classification accuracy.

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4 Current Approaches and Developments

4.1 Corpus-based supervised classification: the Citation-Integrity study

Sarol et al. (2024) constructed the first purpose-built corpus for this task: 285 discussion sections drawn from as many biomedical publications, comprising 4,182 annotated citations, each labeled ACCURATE, INACCURATE, or IRRELEVANT with respect to its cited source. They developed both a rule-based method and supervised models (support vector machines and two deep-learning variants), and separately evaluated few-shot in-context learning with LLMs. Their best-performing pipeline — combining a BM25 retrieval step, a MonoT5 neural re-ranker to select the top-20 candidate evidence sentences, and a fine-tuned MultiVerS model for the final accuracy classification — achieved 0.59 micro-F1 and 0.52 macro-F1. A GPT-4 in-context-learning approach performed somewhat better overall (0.65 micro-F1) and was particularly strong at correctly identifying *accurate* citations, but lagged considerably in identifying *erroneous* ones (0.45 macro-

F1), reflecting the class imbalance and subtlety inherent to quotation errors. The authors conclude that citation quotation errors are "often subtle," and that current NLP models still find them challenging to detect reliably — a caution echoed throughout this literature.

4.2 Replicating Greenberg's citation-distortion analysis with NLP

Sarol, Schneider, and Kilicoglu (2025) undertook a direct test of whether NLP — including LLMs — could automatically reproduce Greenberg's (2009) manual, twelve-year analysis of the inclusion-body-myositis/β-amyloid citation network. They compared a non-LLM pipeline (built from PubMed metadata, MeSH terms, and adapted tools such as MultiVerS) against an LLM pipeline using carefully crafted GPT-4o prompts, across four constituent tasks: (1) classifying each of the 242 papers by publication type (myositis review, animal/cell-culture model, primary data, or other); (2) detecting the stance (supportive, critical, or neutral) of each of 808 citation instances toward the central claim; (3) classifying citation accuracy to detect diversion, transmutation, and dead-end invention; and (4) applying scientific claim verification to detect "title invention," where a paper's title reports results not actually supported in its body text.

Results varied sharply by task. Publication-type classification performed similarly across both approaches (71% accuracy non-LLM vs. 69% LLM), with the non-LLM, metadata-driven method benefiting from strong recall on review-paper identification. Stance detection was the standout success: the LLM approach reached 90% overall accuracy (F1 = 0.96 for the dominant "supportive" class), far outperforming both a repurposed Fake News Challenge stance classifier (40% accuracy) and a citation-sentiment model that had not been given the claim itself (7% accuracy) — the authors show that supplying the claim alongside the citation context, rather than the context alone, was the decisive factor (89% vs. 13% accuracy in a controlled comparison). Building on this, the study successfully replicated Greenberg's amplification finding, correctly identifying all four key review papers through which 95% of citation paths flowed.

By contrast, citation *invention* proved far harder to detect reliably. For citation diversion (4 true cases in the dataset), both approaches achieved only modest accuracy (0.71 non-LLM, 0.75 LLM) and low recall (0.25 and 0.50, respectively), while flagging over 200 citation contexts each as positive — a substantial overprediction problem, with only 51 of these flagged instances overlapping between the two methods. Dead-end citation detection performed somewhat better (0.67 vs. 0.80 accuracy; 0.44 vs. 0.56 recall) but still generated well over one hundred false positives. Citation transmutation, attempted only with the LLM approach, achieved 41% accuracy and 65% recall for the 17 true cases, but with 485 total flagged instances — again indicating extensive overprediction and substantial confusion among invention subtypes. Finally, neither approach detected the single true instance of title invention in the dataset. The authors attribute much of this difficulty to the requirement of full-document input (either of the citing paper itself, for primary-data and title-invention classification, or of the cited article, for invention detection), noting that tasks relying only on short inputs — such as the roughly two-sentence claim-plus-citation-context pairs used for stance detection — performed markedly better than tasks requiring full-text reasoning.

4.3 Citation-quotation-error detection with retrieval-augmented LLMs

Complementing these biomedical, network-focused studies, Zhang and Abernethy (2024) evaluated GPT-family models on a general-domain, expert-annotated dataset of statement–reference pairs, framing the task as a binary judgment of whether a citing statement is "fully substantiated" or only "partially substantiated" by its reference. Varying the amount of retrieved reference context supplied to the model, they found that LLMs could detect erroneous citations even with limited context and without any task-specific fine-tuning — a promising signal for lightweight, deployable citation-checking tools, though, as with the studies above, performance on the harder "partially substantiated" cases (which correspond to Mogull's "minor" quotation errors) remained more limited than on cases of outright contradiction or irrelevance.

4.4 Complementary infrastructure: citation function and sentiment as pre-processing

Because outright verification of a citation's accuracy is computationally and semantically demanding, several current systems use citation function and sentiment classification (Teufel et al., 2006; Cohan et al., 2019; Kilicoglu et al., 2019) as a lightweight pre-processing filter — for example, restricting expensive full-text accuracy verification to citations whose function suggests a substantive evidentiary claim (rather than, say, a citation acknowledging general background) or whose sentiment suggests a strongly supportive or critical stance. This modular design mirrors the claim-detection stage of the general automated fact-checking pipeline (Guo et al., 2022) and offers a practical way to scale citation verification to the volume of citations found in a typical literature review.

4.5 A note on generalization across domains

Almost all corpus-based work reviewed here is drawn from biomedicine, reflecting both the acute real-world stakes of medical misinformation (Leung et al., 2017) and the availability of structured metadata (MeSH terms, publication types) that assist non-LLM pipelines (Sarol et al., 2025; Cohen et al., 2021). Jahan et al. (2024) confirm more broadly that LLMs perform unevenly across benchmark biomedical text-processing tasks, underscoring that strong performance on general NLP benchmarks does not guarantee strong performance on the specialized, evidence-grounded reasoning that citation verification demands.

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5 Research Challenges and Limitations

Full-text access and extraction quality. Verifying whether a citation's claim is substantiated requires the full text of the cited source, not merely its abstract — yet full-text access is frequently restricted by publisher paywalls, and even where text is available, PDF-to-text conversion introduces artifacts (incorrectly merged or split paragraphs, garbled special characters) that degrade downstream model performance (Meuschke et al., 2023; Sarol et al., 2025).

Subtlety and severity gradation. Mogull's (2017) distinction between "major" errors (contradiction, irrelevance) and "minor" errors (oversimplification, overgeneralization) maps onto a real difficulty for automated systems: models trained or prompted to detect binary accuracy consistently perform better on gross contradictions than on graded, rhetorically subtle distortions, which may require modeling implicature and hedging language rather than surface-level semantic overlap.

Overprediction and low precision. A recurring finding across studies is that both LLM and non-LLM pipelines tend to substantially overpredict positive (distortion) labels relative to the small number of true positive cases (Sarol et al., 2025), which is especially problematic for practical deployment: a tool that floods editors or authors with false alarms will not be trusted or adopted, regardless of its recall.

Disambiguating invention subtypes. Citation diversion, transmutation, and dead-end citation frequently overlap in model predictions (Sarol et al., 2025), suggesting either that current prompt or feature designs cannot cleanly separate these categories, or that the underlying phenomena genuinely sit on a continuum that resists crisp categorical boundaries.

Domain and corpus scale limitations. The largest annotated resource to date — Sarol et al.'s (2024) 4,182-citation biomedical corpus — remains modest by NLP standards, and virtually all empirical work reviewed here is confined to biomedicine, leaving open how well these methods transfer to the humanities, social sciences, or physical sciences, where argumentation styles and citation conventions differ substantially.

Distinguishing legitimate paraphrase from distortion. Because scientific writing routinely condenses, simplifies, or recontextualizes prior findings for a new audience or argument, an automated system must distinguish acceptable summarization from a genuine misrepresentation — a judgment that even trained human annotators (as in Greenberg's original manual analysis) required considerable domain expertise to make.

Network-level phenomena resist purely pairwise verification. Citation bias and amplification are properties of an entire citation network, not of any single citing–cited pair; detecting them automatically requires first solving several prerequisite subtasks (publication-type classification, primary-data identification, stance detection) with sufficient accuracy that errors do not compound when aggregated — a requirement not yet reliably met, as reflected in the persistently poor F1 scores (below 0.55 in Sarol et al., 2025) for primary-data-paper identification.

Computational and context-length costs. Feeding full-text documents to LLMs is expensive and often exceeds practical context windows, motivating retrieval-augmented approaches (Lewis et al., 2020) whose own retrieval step introduces an additional source of error if relevant evidence is not surfaced.

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6 Research Gaps and Future Directions

Larger, multi-domain annotated benchmarks. Extending citation-accuracy corpora such as Sarol et al.'s (2024) beyond biomedicine — into fields with different citation norms and rhetorical conventions — would clarify how much of current systems' performance is domain-specific versus generalizable, and would allow the field to benchmark progress the way FEVER (Thorne et al., 2018) and SciFact (Wadden et al., 2020) have done for general and scientific claim verification, respectively.

Explainable, rationale-producing verification. Approaches such as ProoFVer (Krishna et al., 2022), which produce interpretable natural-logic proof traces, and MultiVerS (Wadden et al., 2022), which extracts supporting rationale sentences, point toward citation-verification tools that do not merely output a binary judgment but show *which* passage in the cited source does or does not support the citing claim — a feature likely to be essential for author and editor trust, given the overprediction problems noted above.

Tighter integration of citation function/intent with accuracy verification. Since not all citations make an evidentiary claim requiring full verification (Teufel et al., 2006; Cohan et al., 2019), future pipelines could route citations dynamically: applying lightweight function/sentiment classifiers first, and reserving expensive full-text verification for citations whose function suggests a substantive empirical claim, improving both computational efficiency and precision.

Retrieval-augmented and long-context modeling. Continued development of retrieval methods tailored to scientific text (building on the BM25-plus-neural-reranking pipelines used by Sarol et al., 2024) and of models natively capable of reasoning over long documents could reduce the performance gap currently observed between short-input tasks (e.g., stance detection) and full-text-dependent tasks (e.g., invention detection).

Automated, scalable network-level distortion analysis. Sarol et al. (2025) explicitly frame their replication of Greenberg's manual study as a first step toward a fully automated pipeline capable of identifying all literature relevant to an arbitrary claim, extracting citation contexts at scale, and flagging bias and amplification patterns without requiring a pre-existing, manually curated citation network — a natural and consequential extension given that the original Greenberg network is now nearly two decades old and has likely grown substantially.

Human-in-the-loop and editorial integration. Given current systems' tendency toward overprediction, near-term practical value may come less from fully automated flagging and more from decision-support tools that rank citations by estimated risk of distortion for human reviewers — analogous to how statcheck (Nuijten & Polanin, 2020) has been integrated into peer-review workflows for statistical-reporting checks, with documented reductions in error rates once adopted.

Unifying citation-existence and citation-interpretation verification. As LLM-generated scholarly text becomes more common, tools that jointly check whether a citation exists (Misra & Udandarao, 2026) and whether its claimed content is accurate would provide more complete protection against citation-based misinformation than either check alone, motivating research on combined verification pipelines and shared benchmarks spanning both problem types.

Cross-lingual and interdisciplinary extension. Nearly all cited empirical work is in English-language biomedical literature; extending both annotated resources and models to other languages and disciplines remains an open and largely unaddressed direction.

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7 Conclusion

Citations that point to real, legitimately published sources but misrepresent what those sources report constitute a distinct and consequential category of scientific misinformation — one that existence-checking tools cannot catch and that has, in documented cases, contributed to the durable acceptance of unsubstantiated clinical claims (Greenberg, 2009; Leung et al., 2017). Recasting citation verification as a claim-evidence problem, in the tradition of automated fact-checking (Thorne et al., 2018; Guo et al., 2022) and scientific claim verification (Wadden et al., 2020, 2022), has enabled real progress: automated systems now detect citation stance with high accuracy and can replicate network-level amplification patterns that once required years of manual analysis (Sarol et al., 2025). At the same time, the subtler forms of citation invention — diversion, transmutation, and dead-end citation — remain difficult to detect reliably, with both LLM-based and traditional pipelines showing persistent overprediction and limited recall on the rarer, more nuanced cases (Sarol et al., 2024, 2025; Zhang & Abernethy, 2024). Closing this gap will likely require larger and more diverse annotated corpora, retrieval and long-context modeling advances, explainable verification methods that expose their evidentiary reasoning, and thoughtful integration into the editorial and peer-review processes where such tools can do the most good. As the volume of scientific literature — and of LLM-assisted scholarly writing — continues to grow, automated detection of misinterpreted but genuine citations is likely to become an increasingly necessary complement to, rather than a substitute for, careful human reading of the sources we cite.

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